Unit: Unit 3: Functions
Topic: Inverse Relations and Functions

Name: ____________
Date: ________

Quiz (Carolina Reaper)

Q1. Which of the following is a characteristic of an inverse function?
(A) It must be a linear function
(B) It must be a quadratic function
(C) It must be one-to-one
(D) It must be a polynomial function
(E) It must be a rational function

Q2. Find the inverse of $f(x) = 2x + 1$
(A) $f^{-1}(x) = \frac{x-1}{2}$
(B) $f^{-1}(x) = \frac{x+1}{2}$
(C) $f^{-1}(x) = 2x - 1$
(D) $f^{-1}(x) = \frac{x}{2} - 1$
(E) $f^{-1}(x) = \frac{x}{2} + 1$

Q3. If $f(x) = x^2$, does f have an inverse?
(A) Yes, $f^{-1}(x) = \sqrt{x}$
(B) Yes, $f^{-1}(x) = -\sqrt{x}$
(C) No, f is not one-to-one
(D) No, f is not onto
(E) Maybe, it depends on the domain

Q4. What is the composition of $f(x) = x^2$ and its inverse $f^{-1}(x) = \sqrt{x}$?
(A) $f(f^{-1}(x)) = x$
(B) $f(f^{-1}(x)) = x^2$
(C) $f(f^{-1}(x)) = \sqrt{x}$
(D) $f(f^{-1}(x)) = -x$
(E) $f(f^{-1}(x)) = x + 1$

Q5. Given $f(x) = \frac{1}{x}$, find $f^{-1}(x)$
(A) $f^{-1}(x) = \frac{1}{x}$
(B) $f^{-1}(x) = x$
(C) $f^{-1}(x) = \frac{1}{x^2}$
(D) $f^{-1}(x) = x^2$
(E) $f^{-1}(x) = -x$

Q6. What is the domain of the inverse of $f(x) = x^2$?
(A) $[0, \infty)$
(B) $(0, \infty)$
(C) $(-\infty, 0]$
(D) $(-\infty, \infty)$
(E) $[0, 1)$

Q7. If $f(x) = |x|$, does f have an inverse?
(A) Yes, $f^{-1}(x) = x$
(B) Yes, $f^{-1}(x) = -x$
(C) No, f is not one-to-one
(D) No, f is not onto
(E) Maybe, it depends on the domain

Q8. Given $f(x) = 3x - 2$, find the inverse of f and verify using composition
(A) $f^{-1}(x) = \frac{x+2}{3}$ and $f(f^{-1}(x)) = x$
(B) $f^{-1}(x) = \frac{x+2}{3}$ and $f(f^{-1}(x)) = x + 1$
(C) $f^{-1}(x) = \frac{x-2}{3}$ and $f(f^{-1}(x)) = x$
(D) $f^{-1}(x) = \frac{x-2}{3}$ and $f(f^{-1}(x)) = x - 1$
(E) $f^{-1}(x) = \frac{x}{3} + 2$ and $f(f^{-1}(x)) = x$

Open Ended Questions (FRQ)

Q9. Find the inverse of $f(x) = x^3 + 2$ and verify using composition. Show all steps and explain your reasoning

Q10. A company's profit P (in dollars) is given by the function $P(x) = 200x - 1000$, where x is the number of units sold. Find the inverse function and interpret its meaning in the context of the problem. Show all steps and explain your reasoning

Chef's Tips

- Review the concept of one-to-one functions
- Provide extra practice with finding inverses algebraically
- Emphasize the importance of verifying inverses using composition